
EECS 16A Designing Information Devices and Systems I

Homework 8

Submission Format

Your homework submission should consist of **one** file.

- `hw8.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned).

Submit the file to the appropriate assignment on Gradescope.

1. Midsemester Survey

Please fill out the midsemester survey. The link is given below:

https://docs.google.com/forms/d/e/1FAIpQLSchuTtDXMXqy-uwZdmRwh_3a8UbTjViyfXDILSv7g1PWw_pTg/viewform

Attach a screenshot of the confirmation page to your submission of this homework.

2. Reading Assignment

For this homework, please read Note 15: Section 15.3 to learn about superposition, a concept that can help to simplify circuit analysis. Also read Note 15: Sections 15.4-15.8, which will explain the ideas of finding the equivalent resistance and Thévenin and Norton equivalent circuits. Please also reread Note 14 to review the idea of the 2D touchscreen. You are always encouraged to read beyond this as well.

- (a) As a part of superposition, you need to zero out independent sources. What circuit elements are equivalent to a zeroed voltage source and zeroed current source, respectively?
- (b) If you connect three resistors (each with value R) in series, what will be the equivalent resistance? What happens if you connect these resistors in parallel?
- (c) What is the difference between the Thévenin and Norton equivalent circuits? In particular, which part of the I - V curve does each equivalent circuit correspond to?

3. Measuring Voltage and Current

(Contributors: Ava Tan, Aviral Pandey, Craig Schindler, Taehwan Kim, Urmita Sikder, Vijay Govindarajan, Wahid Rahman)

Learning Goal: *The objective of this problem is to provide a deeper understanding of current and voltage measurement processes. It will also help you to understand how the electrical parameters of a measurement tool can affect the measurement precision.*

In order to measure quantities such as voltage and current, engineers use voltmeters and ammeters. A simple model of a voltmeter is a resistor with a very high resistance, R_{VM} . **The voltmeter measures the voltage across the resistance R_{VM} .** The measured voltage is then relayed to a microprocessor (such as the MSP430 microprocessor, which will be used in lab).

This model of a voltmeter is shown in Figure 1. Let us explore what happens when we connect this voltmeter to various circuits to measure voltages.

Throughout this problem assume $R_{VM} = 1M\Omega$. Recall that the SI prefix M or Mega is 10^6 .

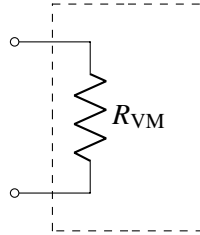


Figure 1: Our model of a voltmeter, $R_{VM} = 1M\Omega$

- (a) Suppose we wanted to measure the voltage across R_2 (v_{out}) produced by the voltage divider circuit shown in Figure 2 on the left. The circuit on the right in Figure 2 shows how we would connect the voltmeter across R_2 . Assume $R_1 = 100\Omega$ and $R_2 = 200\Omega$.

First calculate the value of v_{out} . Then calculate the voltage the voltmeter would measure, i.e. v_{meas} .

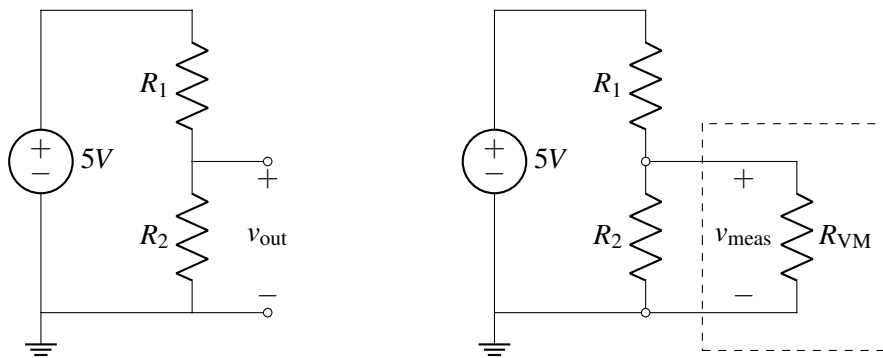


Figure 2: *Left*: Circuit without the voltmeter connected. *Right*: Voltmeter measuring voltage across R_2 .

- (b) Repeat part (a), but now $R_1 = 10M\Omega$ and $R_2 = 10M\Omega$. Is this particular voltmeter still a good tool to measure the output voltage? Justify why or why not. (Notice that a *good* voltmeter should not significantly affect the value of voltages in a circuit by its presence.)
- (c) Now suppose we are working with the same circuit as in part (a), but we know that $R_2 = R_1$. What is the maximum value of R_1 such that $v_{out} - v_{meas} \leq 0.1 \cdot v_{out}$ (i.e. v_{meas} is only 10% smaller than v_{out})?
- (d) We can make **an ammeter** and measure the current through an element, using the combination of our voltmeter and an additional resistor R_x . The circuit shown in Figure 3 encompassed by the dashed box can work as an ammeter, where $R_x = 1\Omega$. We insert it in the circuit so that the current we want to measure flows through R_x , and then measure the current as $I_{meas} = \frac{V_{VM}}{R_x}$ where V_{VM} is the voltage across the voltmeter. $R_{VM} = 1M\Omega$ is the same as in previous parts.

In Figure 4, the voltmeter-resistor combo is connected so that the current we want to measure flows through resistor $R_1 = 1k\Omega$. For the circuit on the left, find the current through R_1 without the voltmeter-resistor combo connected (i.e. I_1). Then, for the circuit on the right, find the current measured by the voltmeter-resistor combo when it is connected as an ammeter (i.e. I_{meas}).

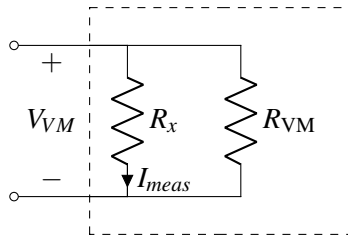


Figure 3: The voltmeter combined with resistor R_x to function as an ammeter (i.e. to measure current), $R_{VM} = 1M\Omega$.

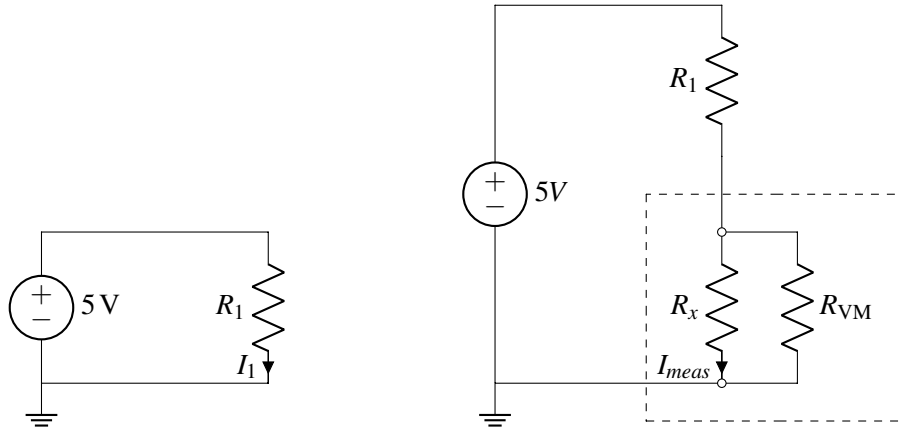


Figure 4: Circuits for part (d). *Left:* Original circuit; *Right:* Circuit with the voltmeter connected as an ammeter.

- (e) **(PRACTICE / NOT GRADED)** What is the minimum value of R_1 that ensures the difference between current measurement (I_{meas}) and the actual value (I_1) is such that $I_1 - I_{meas} \leq 0.1 \cdot I_1$, i.e. stays within $\pm 10\%$ of I_1 ? In other words, find the minimum allowable value for R_1 such that $I_1 - I_{meas} \leq 0.1 \cdot I_1$.

Hint: You can approximate $R_{VM} || R_x \approx R_x$ and $\frac{R_{VM} || R_x}{R_x} \approx 1$.

4. Digital to Analog Converter (DAC)

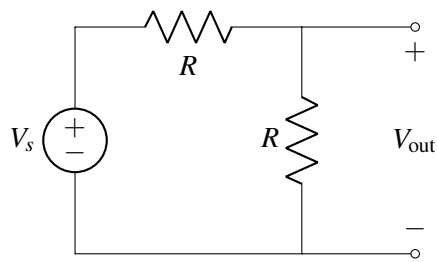
For some outputs, such as audio applications, we need to produce an analog output, or a continuous voltage from 0 to V_s . These analog voltages must be produced from digital voltages, that is sources, that can only be V_s or 0. A circuit that does this is known as a Digital to Analog Converter. It takes a binary representation of a number and turns it into an analog voltage.

The output of a DAC can be represented with the equation shown below:

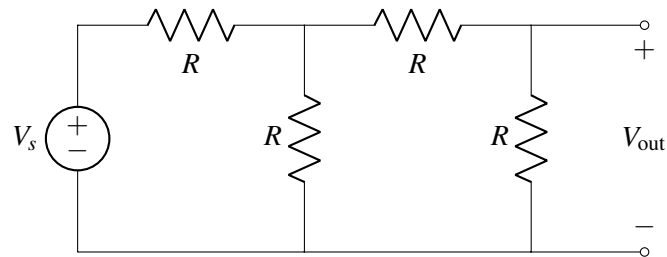
$$V_{out} = V_s \sum_{n=0}^N \frac{1}{2^n} \cdot b_n$$

where each binary digit b_n is multiplied by $\frac{1}{2^n}$.

- (a) We know how to take an input voltage and divide it by 2:



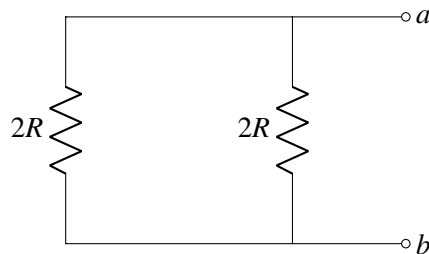
To divide by larger powers of two, we might hope to just “cascade” the above voltage divider. For example, consider:



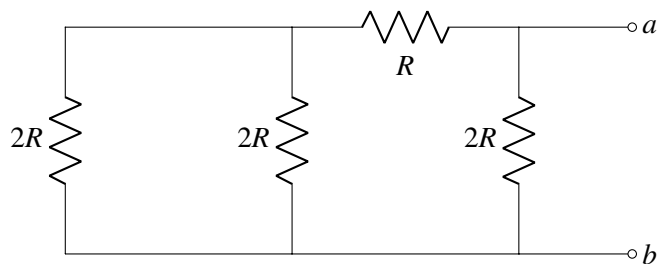
Calculate V_{out} in the above circuit. Is $V_{\text{out}} = \frac{1}{4}V_s$?

- (b) The R - $2R$ ladder, shown below, has a very nice property. For each of the circuits shown below, find the equivalent resistance looking in from points a and b . Do you see a pattern?

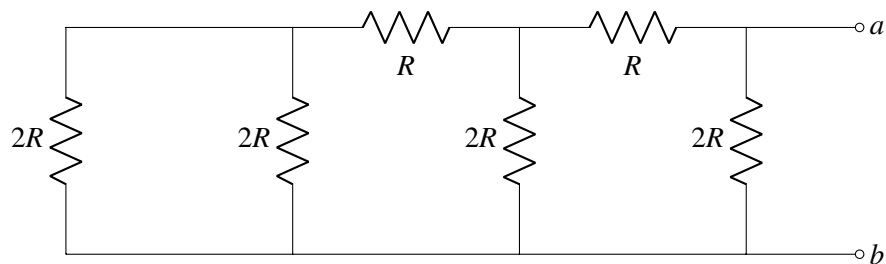
i.



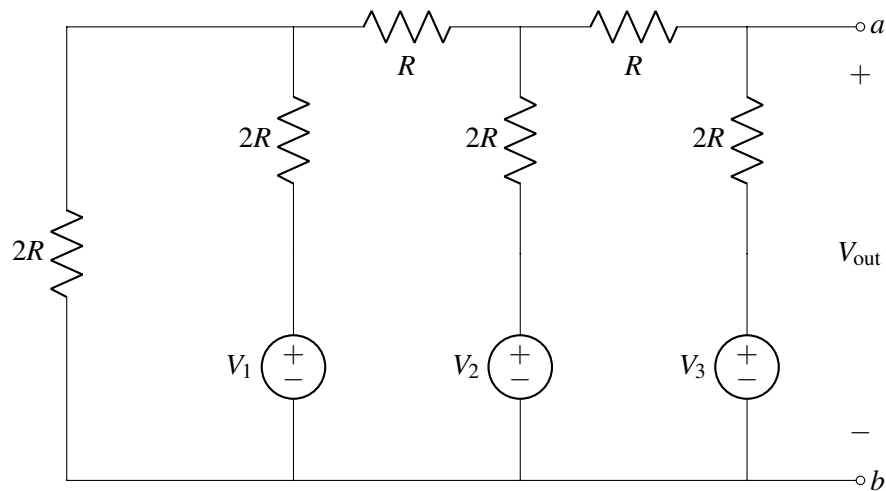
ii.



iii.



- (c) The following circuit is an R - $2R$ DAC. To understand its functionality, use superposition to find V_{out} in terms of each V_k in the circuit.



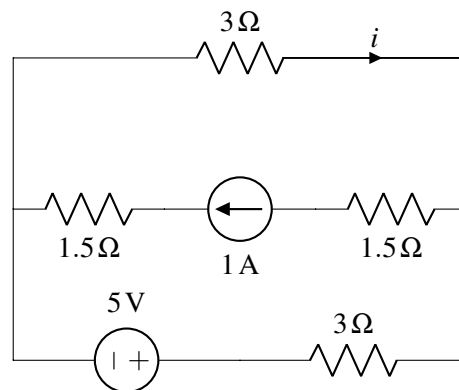
- (d) We've now designed a 3-bit R - $2R$ DAC. What is the output voltage V_{out} if $V_2 = 1\text{ V}$ and $V_1 = V_3 = 0\text{ V}$?

5. Superposition

(Contributors: Avi Pandey, Gireeja Ranade, Karina Chang, Raghav Anand, Panos Zarkos, Sashank Krishnamurthy, Titan Yuan)

Learning Goal: The objective of this problem is to help you practice solving circuits using the principles of superposition.

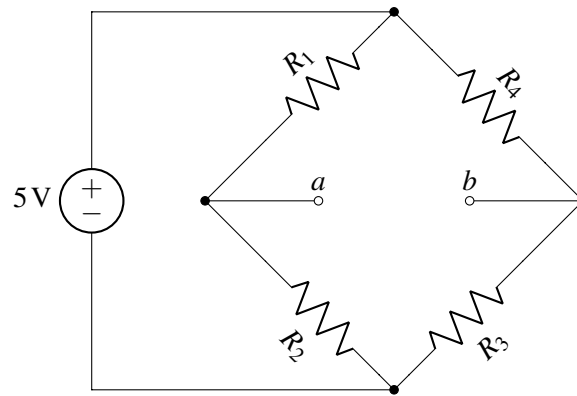
Find the current i indicated in the circuit diagram below using superposition.



6. Wheatstone Bridge

(Contributors: Ava Tan, Panos Zarkos)

A Wheatstone Bridge is a very useful circuit that can be used to help determine unknown resistance values with very high accuracy. This circuit is used in many sensor applications where resistors $R_1 - R_4$ are varying with respect to some external actuation. For example, it can be used to build a strain gauge (https://en.wikipedia.org/wiki/Strain_gauge) or a scale (remember Fred's Fruits from HW 7?). In that case the resistors $R_1 - R_4$ would vary with respect to a strain caused by a force, and the Wheatstone Bridge circuit would translate that variation into a voltage difference across the "bridge" terminals a and b .



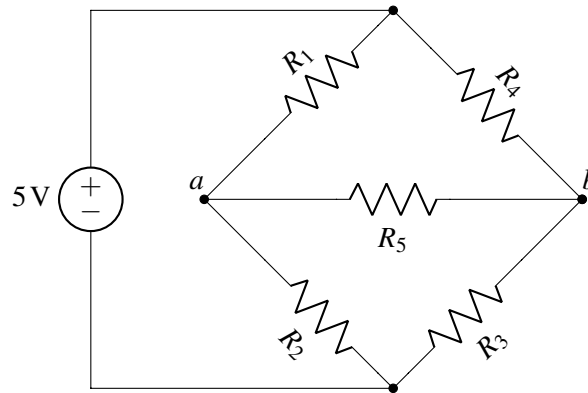
- (a) Assume that $R_1 = 2\text{ k}\Omega$, $R_2 = 2\text{ k}\Omega$, $R_3 = 1\text{ k}\Omega$, $R_4 = 4\text{ k}\Omega$. Calculate the voltage V_{ab} between the two terminals a and b .

Hint: You have analyzed very similar circuits in previous homeworks, in lectures, and in discussions – it may help to redraw the circuit with each branch containing resistors $R_1 - R_4$ using straight vertical wires, rather than diagonal ones.

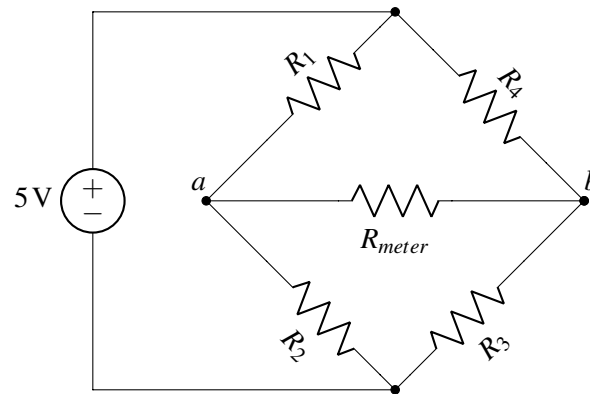
- (b) Now assume that you have added an additional resistor, R_5 , between terminals a and b as shown below. Assume that $R_1 = 1\text{ k}\Omega$, $R_2 = 2\text{ k}\Omega$, $R_3 = 4\text{ k}\Omega$, $R_4 = ?\text{ k}\Omega$, $R_5 = 4\text{ k}\Omega$. In the process of constructing this circuit, you notice that you forgot to write down the value of R_4 !

However, you notice something very curious about your Wheatstone Bridge circuit: there is no current flowing through resistor R_5 .

Based on this observation, what must the value of resistor R_4 be?



- (c) Next, draw the Thévenin equivalent of the Wheatstone bridge circuit between terminals a and b , using resistor values from part (a).
- (d) Now assume that you are trying to measure the voltage V_{ab} between nodes a and b using a voltmeter, whose resistance is R_{meter} , so you end up with the circuit below.

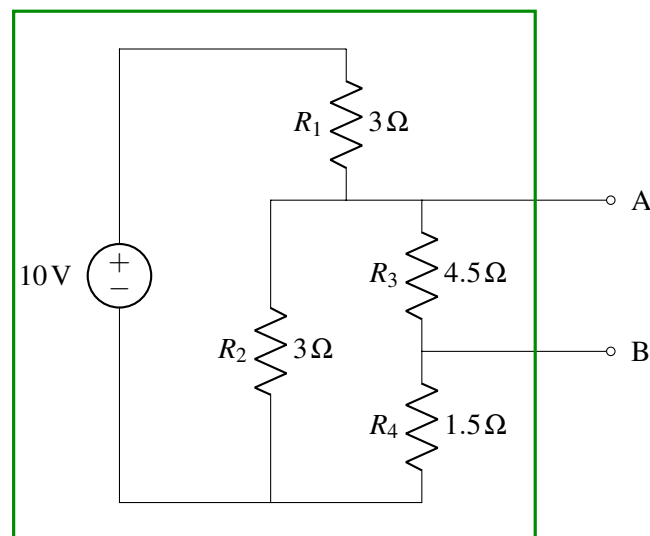


Unfortunately, your voltmeter is far from ideal, so $R_{meter} = 4k\Omega$. Is the voltage V_{ab} you found in part (a) equal to the new voltage $V_{R_{meter}}$ across the voltmeter resistor? Why or why not? Calculate the current $I_{R_{meter}}$ through the voltmeter resistor and the voltage $V_{R_{meter}}$ across the meter resistor. Again, use resistor values from part (a) and $R_{meter} = 4k\Omega$.

Hint: Consider the equivalent circuit you found in part (c).

7. Thévenin and Norton Equivalent Circuits

Find the Thévenin and Norton equivalent circuits seen from terminals A and B.



8. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.